1.58 Equations of Planes - Scalar Product Equations

Worked Solutions

1)

$$\left(\mathbf{r} - \begin{pmatrix} -1 \\ 8 \\ 2 \end{pmatrix}\right) \cdot \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix} = 0.$$

2)

$$\cos\theta = \frac{\begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ -1 \\ 2 \end{pmatrix}}{\sqrt{2^2+0^2+1^2} \times \sqrt{6^2+(-1)^2+2^2}}.$$

$$\Rightarrow \cos\theta = \frac{2 \times 6 + 0 \times (-1) + 1 \times 2}{\sqrt{5} \times \sqrt{41}}.$$

$$\Rightarrow \cos\theta = \frac{14}{\sqrt{205}} \Rightarrow \theta = 12.094\ldots^\circ.$$

Acute angle is 12.1° (1 DP).

3)

$$\left(\begin{pmatrix} 3 \\ \frac{8}{5} \\ 7 \end{pmatrix} - \begin{pmatrix} 5 \\ 2 \\ -3 \end{pmatrix}\right) \cdot \begin{pmatrix} -6 \\ 5 \\ -1 \end{pmatrix} = \begin{pmatrix} -2 \\ -\frac{2}{5} \\ 10 \end{pmatrix} \cdot \begin{pmatrix} -6 \\ 5 \\ -1 \end{pmatrix}.$$

$$= (-2) \times (-6) + \left(-\frac{2}{5}\right) \times 5 + 10 \times (-1).$$

$$= 12 - 2 - 10 = 0.$$

So $\left(3, \frac{8}{5}, 7\right)$ lies on the plane Π.

4)

$$\begin{pmatrix} 2 \\ -5 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} -9 \\ 8 \\ -16 \end{pmatrix} = 2 \times (-9) + (-5) \times 8 + (-4) \times (-16) = -18 - 40 + 64 = 6.$$

So $(2, -5, -4)$ lies on the plane Π.

5 a)

$$\mathbf{r} \cdot \mathbf{i} = 2 \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 2 \Rightarrow x = 2.$$

5 b)

$$\mathbf{r} \cdot \mathbf{j} = 0 \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0 \Rightarrow y = 0.$$

5 c)

$$\mathbf{r} \cdot \mathbf{k} = -3 \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = -3 \Rightarrow z = -3.$$

5 d)

$$\mathbf{r} \cdot (3\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) = 9 \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \\ -3 \end{pmatrix} = 9 \Rightarrow 3x + 2y - 3z = 9.$$

5 e)

$$\mathbf{r} \cdot (10\mathbf{i} - 50\mathbf{j} + 25\mathbf{k}) = 30 \Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 10 \\ -50 \\ 25 \end{pmatrix} = 30 \Rightarrow 10x - 50y + 25z = 30.$$
$$\Rightarrow 2x - 10y + 5z = 6.$$

6)

The plane has equation $2x - 3y - 10z = d$.

$$d = 2(3) - 3(-6) - 10(8) = -56.$$
$$\therefore 2x - 3y - 10z = -56.$$
$$\Rightarrow \mathbf{r} \cdot \begin{pmatrix} 2 \\ -3 \\ -10 \end{pmatrix} = -56.$$

7 a)

$$\cos\theta = \frac{\begin{pmatrix} -1 \\ -2 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ 10 \end{pmatrix}}{\sqrt{(-1)^2 + (-2)^2 + 4^2} \times \sqrt{3^2 + 1^2 + 10^2}}.$$
$$\Rightarrow \cos\theta = \frac{(-1) \times 3 + (-2) \times 1 + 4 \times 10}{\sqrt{21} \times \sqrt{110}}.$$
$$\Rightarrow \cos\theta = \frac{35}{\sqrt{2310}} \Rightarrow \theta = 43.262\ldots^\circ.$$

Acute angle is 43.3° (1 DP).

7 b)

$$\cos\theta = \frac{\begin{pmatrix}-5\\-1\\-9\end{pmatrix}\cdot\begin{pmatrix}16\\2\\-5\end{pmatrix}}{\sqrt{(-5)^2+(-1)^2+(-9)^2}\times\sqrt{16^2+2^2+(-5)^2}}.$$

$$\Rightarrow \cos\theta = \frac{(-5)\times 16+(-1)\times 2+(-9)\times(-5)}{\sqrt{107}\times\sqrt{285}}.$$

$$\Rightarrow \cos\theta = -\frac{37}{\sqrt{30495}} \Rightarrow \theta = 102.232\ldots^\circ.$$

Acute angle is $180^\circ - 102.232\ldots^\circ = 77.8^\circ$ (1 DP).

8)

$$\cos\theta = \frac{\begin{pmatrix}3\\-2\\-10\end{pmatrix}\cdot\begin{pmatrix}0\\0\\1\end{pmatrix}}{\sqrt{3^2+(-2)^2+(-10)^2}\times\sqrt{0^2+0^2+1^2}}.$$

$$\Rightarrow \cos\theta = \frac{3\times 0+(-2)\times 0+(-10)\times 1}{\sqrt{113}\times\sqrt{1}}.$$

$$\Rightarrow \cos\theta = -\frac{10}{\sqrt{113}} \Rightarrow \theta = 160.172\ldots^\circ.$$

Acute angle is $180^\circ - 160.172\ldots^\circ = 19.8^\circ$ (1 DP).

9 a)

$$\begin{pmatrix}2\\3\\1\end{pmatrix}\cdot\begin{pmatrix}8\\-1\\1\end{pmatrix} = 2\times 8+3\times(-1)+1\times 1 = 14.$$

$$\Pi_1 : \mathbf{r}\cdot\begin{pmatrix}8\\-1\\1\end{pmatrix} = 14.$$

9 b)

$$\begin{pmatrix}3\\9\\2\end{pmatrix}\cdot\begin{pmatrix}1\\-3\\-2\end{pmatrix} = 3\times 1+9\times(-3)+2\times(-2) = -28.$$

$$\Pi_2 : \mathbf{r}\cdot\begin{pmatrix}1\\-3\\-2\end{pmatrix} = -28.$$

9 c)

$$\cos\theta = \frac{\begin{pmatrix}8\\-1\\1\end{pmatrix}\cdot\begin{pmatrix}1\\-3\\-2\end{pmatrix}}{\sqrt{8^2+(-1)^2+1^2}\times\sqrt{1^2+(-3)^2+(-2)^2}}.$$
$$= \frac{8\times 1+(-1)\times(-3)+1\times(-2)}{\sqrt{66}\times\sqrt{14}}.$$
$$= \frac{9}{\sqrt{924}} = \frac{3\sqrt{231}}{154}.$$

10 a)

$$\begin{pmatrix}800t^2+540t-99\\300-20t\\800t^2+620t-133\end{pmatrix}\cdot\begin{pmatrix}1\\0\\-1\end{pmatrix} = 0.$$
$$\Rightarrow (800t^2+540t-99)\times 1+(300-20t)\times 0+(800t^2+620t-133)\times(-1)=0.$$
$$\Rightarrow 800t^2+540t-99-800t^2-620t+133=0.$$
$$\Rightarrow -80t+34=0 \Rightarrow t=\frac{17}{40}=0.425 \text{ seconds.}$$

10 b)

$$\mathbf{v}|_{t=\frac{17}{40}} = \begin{pmatrix}1600\times\frac{17}{40}+540\\-20\times\frac{17}{40}\\1600\times\frac{17}{40}+620\end{pmatrix} = \begin{pmatrix}1220\\-\frac{17}{2}\\1300\end{pmatrix}.$$

10 c)

$$\cos\theta = \frac{\begin{pmatrix}1220\\-\frac{17}{2}\\1300\end{pmatrix}\cdot\begin{pmatrix}1\\0\\-1\end{pmatrix}}{\sqrt{1220^2+\left(-\frac{17}{2}\right)^2+1300^2}\times\sqrt{1^2+0^2+(-1)^2}}.$$
$$\Rightarrow \cos\theta = \frac{1220\times 1+\left(-\frac{17}{2}\right)\times 0+1300\times(-1)}{\sqrt{3178472.25}\times\sqrt{2}}.$$
$$\Rightarrow \cos\theta = -\frac{80}{\sqrt{6356944.5}} \Rightarrow \theta = 91.818\ldots^\circ.$$

So the acute angle between the particle and the normal vector to the plane is $180^\circ - 91.818\ldots^\circ = 88.181\ldots^\circ$. So the acute angle between the particle and the plane is $90^\circ - 88.181\ldots^\circ = 1.8^\circ$ (1 DP).

11 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 300 \\ 50 \\ -150 \end{pmatrix} - \begin{pmatrix} 150 \\ 0 \\ -200 \end{pmatrix} = \begin{pmatrix} 150 \\ 50 \\ 50 \end{pmatrix}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix} 400 \\ 150 \\ -300 \end{pmatrix} - \begin{pmatrix} 150 \\ 0 \\ -200 \end{pmatrix} = \begin{pmatrix} 250 \\ 150 \\ -100 \end{pmatrix}.$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 150 \\ 50 \\ 50 \end{pmatrix} \times \begin{pmatrix} 250 \\ 150 \\ -100 \end{pmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 150 & 50 & 50 \\ 250 & 150 & -100 \end{vmatrix}.$$

$$= (50 \times (-100) - 50 \times 150)\mathbf{i} - (150 \times (-100) - 50 \times 250)\mathbf{j} + (150 \times 150 - 50 \times 250)\mathbf{k}.$$

$$= -12500\mathbf{i} + 27500\mathbf{j} + 10000\mathbf{k}.$$

The plane has equation $-12500x + 27500y + 10000z = d$.

$$d = -12500(150) + 27500(0) + 10000(-200) = -3875000.$$

$$\therefore -12500x + 27500y + 10000z = -3875000.$$

$$\Rightarrow -5x + 11y + 4z = -1550.$$

So $\Pi : \mathbf{r} \cdot \begin{pmatrix} -5 \\ 11 \\ 4 \end{pmatrix} = -1550.$

11 b)

$$\cos\theta = \frac{\begin{pmatrix} -5 \\ 11 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}{\sqrt{(-5)^2 + 11^2 + 4^2} \times \sqrt{0^2 + 0^2 + 1^2}}.$$

$$\Rightarrow \cos\theta = \frac{(-5) \times 0 + 11 \times 0 + 4 \times 1}{\sqrt{162} \times \sqrt{1}}.$$

$$\Rightarrow \cos\theta = \frac{4}{\sqrt{162}} = \frac{2\sqrt{2}}{9} \Rightarrow \theta = 71.683\ldots^\circ.$$

Acute angle is 71.7° (1 DP).